

Job Proposal | Employee Turnover Prediction

Proposed approach

I would approach employee-retention analytics as a complete data science workflow: clarify the decision, audit the data, explore patterns, build an interpretable baseline, compare machine-learning models, evaluate them against the business objective, explain the strongest signals, and communicate the results clearly.

What this case study demonstrates

- 14,999 employee records analyzed.
- 3,008 duplicate records identified and removed.
- Workload and project-related patterns investigated.
- Logistic regression used as a baseline.
- Random Forest used for nonlinear classification.
- An overworked feature engineered from average monthly hours.
- Final saved holdout result: 96.2% accuracy, 90.4% recall, 88.7% F1-score, 93.8% AUC.
- Feature importance translated into practical HR investigation areas.

How I would apply this to a client project

- 1 Clarify the business decision and the cost of missed cases versus false alerts.
- 2 Audit the source data and define a reproducible cleaning pipeline.
- 3 Explore distributions, segment differences, and data-quality issues.
- 4 Build an interpretable baseline before adding more complex models.
- 5 Engineer features only when they have a clear analytical purpose.
- 6 Compare models using metrics that match the business objective.
- 7 Explain the model and identify the strongest predictive signals.
- 8 Deliver a technical notebook plus a concise decision-oriented report.

Deliverables

Cleaned analysis workflow, exploratory visualizations, baseline and machine-learning models, evaluation and interpretation, reproducible notebook, executive summary, practical recommendations, and next steps.

Modeling principle

For HR use cases, prediction should support investigation and planning rather than automate employment decisions. Model performance should be validated on fresh, representative data before operational use.